CITS4402 Computer Vision Project Report
Face Detection and Matching

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1. Introduction

This report describes the face detection and identity clustering system implemented for the CITS4402 project. The system processes 640×480 colour images (1–4 faces), detects faces, localises three landmarks (right eye, left eye, nose tip), aligns each face to a 125×125 crop, and clusters faces across a folder into unique identities. The pipeline uses MediaPipe BlazeFace and FaceLandmarker when available, with a fully self-contained fallback built on OpenCV Haar cascades and the Hough circle transform (Lab 04). For clustering, 128-D dlib ResNet embeddings (via the `face_recognition` library) feed DBSCAN. The table below shows the headline measured outcomes from the naive starting point to the final implementation; subsequent sections detail each task's development.

1.1 Baseline vs Final (measured on the 12 sample images)

Metric	Naive baseline	Final implementation
Face detection	11/12 images (4.jpg missed)	12/12 images, all counts correct
Landmark detection	61% both-eyes (17/28 faces)	100% both-eyes (26/26 faces)
Landmarks inside box	not measured	100% RE, 100% LE, 100% nose
Alignment success	61%, no similarity transform	100%, exact targets (0 px residual)
Identity clustering	not implemented	eps=0.38, 6 clusters from 26 faces

2. Face Detection (Task 2)

2.1 Development Iterations

Version	Change	Outcome
v1 (baseline)	Haar cascade (`frontalface_default`) only, no skin filter	11/12 images; 4.jpg missed entirely
v2	Added MediaPipe BlazeFace (short-range + full-range) as primary detector; Haar retained as fallback	12/12 images, but false positives: 5 boxes on 11.jpg (small body region), 1 face missed on 12.jpg (Ross failed cascade re-verification)
v3	Added skin-coverage bypass for verification (skin $\geq$ 75% AND box $\geq$ 60 px); cap result at 4 (spec maximum)	Fixed 12.jpg (Ross retained) and 11.jpg (small body box dropped); 12/12 correct
v4	Replaced YCrCb-only skin filter with three-rule majority vote (YCrCb + HSV wrap-around + normalised-RGB)	More tolerant of unusual lighting/darker skin without losing FP rejection
v5 (final)	CLAHE replaces `equalizeHist` for the Haar-fallback grayscale input	12/12 maintained; better local detail for the Haar fallback path

2.2 Final Method

Primary: MediaPipe BlazeFace short-range (230 KB) + full-range (1.1 MB) TFLite models, run on the original and horizontally-flipped image. Results merged with IoU-NMS (threshold 0.28, containment $>$ 72%). Each box expanded 20% up, 10% left/right, 15% down to capture the full head. Capped at 4 by ranking on confidence + skin-coverage score. Fallback when MediaPipe is unavailable: `frontalface_default` + `alt2` Haar cascades, 28% upward expansion, ROI re-verification, profile cascade as last resort.

2.3 Skin-Colour Segmentation (Mandatory)

Three-rule majority vote (Lab 05 technique): a pixel is skin when ≥ 2 of three rules agree — YCrCb (Cr $\in [133, 183]$, Cb $\in [75, 135]$), HSV (H $\in [0, 28] \cup [160, 179]$), normalised RGB (R $>$ G $- 5$, R $>$ B, R > 70). Morphological open + close (5×5 ellipse) before the 20% pixel-ratio threshold.

2.4 Alternatives Considered

- dlib HOG face detector: lower recall on non-frontal faces vs BlazeFace.
- OpenCV YuNet: needs a separate ONNX file not bundled with `opencv-python`.

3. Facial Landmark Detection (Task 3)

3.1 Development Iterations

Version	Change	Outcome
v1 (baseline)	`haarcascade_eye.xml` only	61% both-eyes (17/28); fails on tilted, glasses, occluded faces
v2	Hough circle transform with Lab 04 edge-overlap scoring on upper 58% ROI; `eye_tree_eyeglasses` and directional eye cascades as chained fallbacks; nose tip via intensity minimum of Gaussian-blurred patch	100% both-eyes on all 26 faces
v3	Added MediaPipe FaceLandmarker (iris landmarks 468/473, nose landmark 1) as primary path	100% maintained; sub-pixel iris precision
v4	Per-eye position validation: reject MediaPipe landmarks where any single eye is outside the top 8–60% of the face box, or eyes differ in y by $>$ 20% face height	Caught a Chandler-in-7.jpg case where MediaPipe placed the right eye on the forehead, causing a badly tilted alignment
v5 (final)	Integral-projection profile + symmetry-mirror fallback tiers; CLAHE preprocessing	100% even with MediaPipe disabled (verified by forcing the landmarker off)

3.2 Final Method

Seven-tier fallback chain (each tier only runs if the previous failed): (1) MediaPipe FaceLandmarker on a padded ROI; (2) Hough circles with Lab 04 edge-overlap scoring; (3) `haarcascade_eye_tree_eyeglasses`; (4) directional eye cascades; (5) integral-projection profile (darkest row in the upper face); (6) symmetry mirror if one eye is found; (7) proportional geometric estimate. Nose tip always from intensity minimum in the central 30–70% $\times$ 50–72% region of the face ROI.

3.3 Alternatives Considered

- dlib 68-point shape predictor: 68 MB model file, too large for submission.

4. Facial Alignment (Task 4)

4.1 Development Iterations

Version	Change	Outcome
v1 (baseline)	`getRotationMatrix2D` for rotation, then separate crop + resize. Does not use a similarity transform; landmarks not at target positions	61% success (tied to landmark failures)
v2	`cv2.estimateAffinePartial2D` with LMEDS on all three landmarks	100% success but visible tilt artefact when nose constraint is geometrically inconsistent with eyes (e.g., Monica in 7.jpg, eyes both at y=133 but face aligned with $\sim$20° rotation)
v3	Analytical 2-point similarity transform from the two eye points only	100%, no tilt; eyes exact at (40,40) and (85,40); but nose drifted from (63,70) — flagged by the lab facilitator
v4	Least-squares similarity transform over all three landmarks (closed form via `np.linalg.lstsq` on the 6-equation 4-unknown system)	100%; reduced tilt vs LMEDS but still measured max 9.5 px residual because a 4-DOF model cannot exactly fit 3 arbitrary point pairs
v5 (final)	`cv2.getAffineTransform` with all three landmarks	100%; verified 0 px residual on all 26 faces — every landmark lands precisely at its target coordinate

4.2 Final Method

`cv2.getAffineTransform` takes exactly 3 source$\rightarrow$target pairs and returns the unique 6-DOF affine matrix that maps them exactly. A single `cv2.warpAffine` call with output size (125, 125) simultaneously aligns, crops, resizes, and (where needed) lightly stretches the face so that right_eye $\rightarrow (40, 40)$, left_eye $\rightarrow (85, 40)$, and nose_tip $\rightarrow (63, 70)$ are all exact. `borderMode=BORDER_REPLICATE` fills out-of-bounds pixels, preventing black borders. Landmark dots on the corner thumbnail are drawn at the fixed target coordinates and never move between images.

5. Identity Clustering (Task 5)

5.1 Development Iterations

Version	Change	Outcome
v1 (baseline)	Not implemented in the baseline code	—
v2	128-D dlib ResNet embeddings via `face_recognition` (explicitly permitted by spec); DBSCAN with `min_samples=1` and Euclidean distance	Clusters identities meaningfully; see eps sweep below
v3	Singleton-merge post-processing: merge isolated one-face clusters into the nearest compatible cluster if distance < 0.515 AND no shared source image	Repairs cases where one image of a person has unusual pose/lighting
v4	Upgraded the LBP fallback descriptor from basic 256-bin 5×5 to LBP-u2 7×7 multi-scale (Ahonen et al. 2006): 59 uniform-pattern bins × 49 cells × 2 radii = 5782-D L2-normalised	LBP fallback peak silhouette: 0.04 → 0.11 ($\approx 3\times$ improvement)

5.2 DBSCAN eps Sweep (final code, 26 faces)

eps	Clusters	Silhouette
0.35	10	0.1408
0.38	**6**	**0.1021 ← selected**
0.40	4	0.0954
0.43	1	0.0000
0.45	1	0.0000

eps=0.38 was selected because 6 clusters is closest to the 5-identity ground truth (the 5 sample-image characters), with one slight over-split for a character whose appearance varies significantly between images. eps=0.35 has marginally higher silhouette (0.14) but over-splits to 10 clusters. eps=0.40 collapses two distinct identities. `min_samples=1` ensures every face joins a cluster (no noise points).

5.3 Alternatives Considered

- k-means: cannot be used — number of identities is unknown a priori.
- Agglomerative clustering: rejected because DBSCAN is explicitly named in the spec.

6. Final Results

Measured on the 12 sample images (26 detected faces, all face counts match ground truth):

- Face detection: 12/12 images correct; avg detection $\approx$ 39 ms (MediaPipe path).
- Landmark detection: 100% both eyes; 100% RE, 100% LE, 100% nose inside the face box.
- Alignment: 100% success; 0 px residual on every landmark; all outputs exactly 125×125.
- Clustering: 6 DBSCAN clusters from 26 faces (eps=0.38, silhouette = 0.10).

Known limitations. (1) MediaPipe model files ($\approx$ 4.9 MB total) must accompany the script — the Haar/Hough fallback covers their absence automatically. (2) `face_recognition` (dlib) must be pip-installed for best clustering — the LBP-u2 fallback works without it but with reduced discrimination. (3) eps=0.38 over-splits one identity (6 clusters from 5 characters); eps=0.40 merges two distinct characters, so 6 is the better error.